

Behavioral Health Risk Stratification and Intervention Strategy: A Predictive Analysis.

Business Problem

The State Mental Health Agencies (SMHAs) fund and oversee a broad range of behavioral health services across the United States and its territories including screening, assessment, crisis intervention, and telemedicine. Yearly, millions of individuals benefit from these public funded services, while available resources remain limited relative to demand. Given the high prevalence and complexity of mental illness, service delivery agencies need a scalable and data-drive approach to identify individuals who are most likely to meet the criteria for at least one serious mental illness diagnosis. Early detection would allow agencies to prioritize care coordination, intervention, risk monitoring, and resource allocation for individuals with greater risk.

Objective

The objective of this project was to develop machine learning models capable of predicting serious mental illness (SMI) among adults receiving publicly funded behavioral health care across the United States. The project also seeks to identify the strongest predictors (features) associated with a serious mental illness (SMI) diagnosis and identify meaningful client segments that could support targeted, responsive, and personalized intervention strategies. Ultimately, the analysis was intended to support data-driven service planning and decision-making in the allocation of resources within behavioral health.

Data

This project focuses on the adult population served in 2023, using the open-source mental health client-level data (MH_CLD) available on the SAMHSA website (<https://www.samhsa.gov/data/data-we-collect/mh-cld-mental-health-client-level-data/datafiles/2023>). This data includes reports of the demography, diagnosis, substance use, employment, residential data, and service use of individuals served through the state mental health agencies for the year referred to as feature variables. The data includes one record per individual served, totaling 7,035,641 records. The actual number of individual records used (sample size) in these predictive analytics is 200,000.

Exclusion: American Samoa, Guam, Maine, Marshall Islands, U.S. Virgin Islands, and Federated States of Micronesia reported insufficient data and were excluded from the 2023 report by the Center for Behavioral Health Statistics and Quality, Substance Abuse and Mental Health Services Administration (SAMHSA).

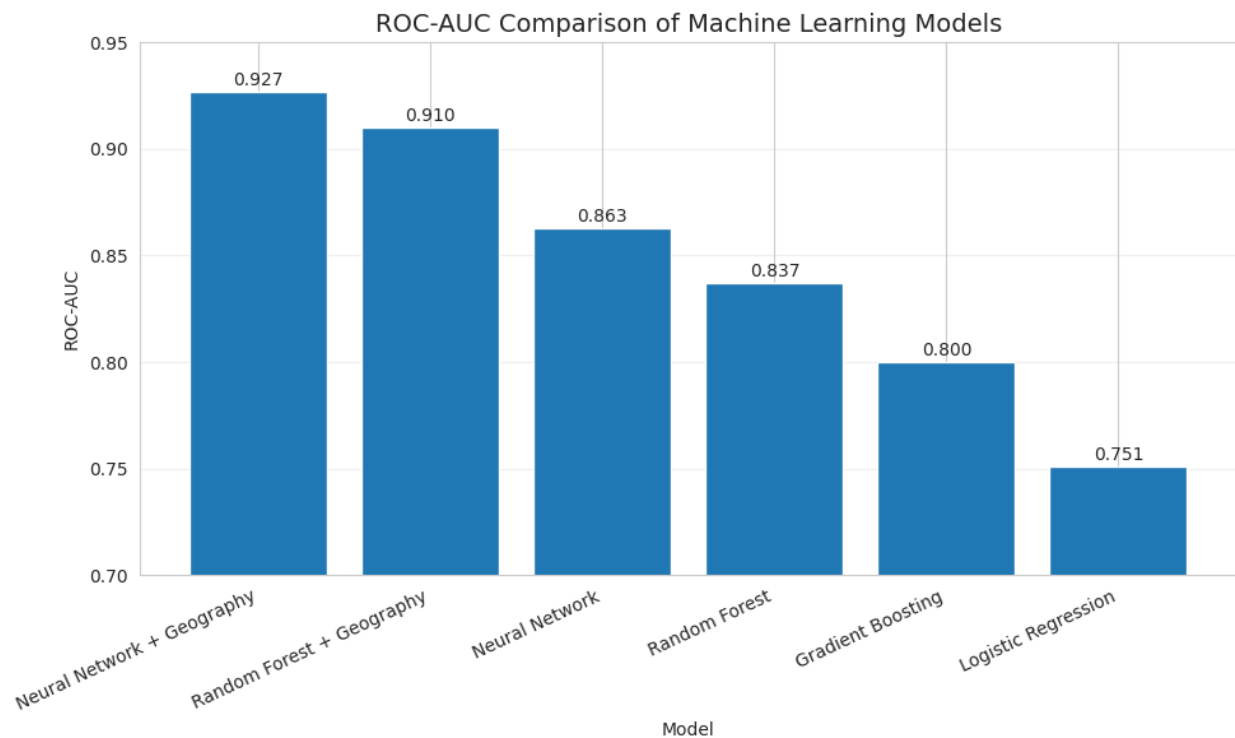
Methodology and Key Findings

The algorithms evaluated in this project include Logistic Regression, Gradient Boosting, Random Forest, and Neural Networks; all supervised machine learning models. Additionally, K-Means Clustering was evaluated as the only unsupervised model included. Among the supervised models, the Neural Network with geographic variables outperformed all other models by achieving an ROC-AUC of 0.927 and an accuracy of 0.844. This suggests that nonlinear machine learning models can efficiently identify clients that have high risks for SMI classification, especially within a geographic context. See Table 1 and Figure 1 for details of each supervised model output.

Table 1: Ranking of all supervised machine learning models

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Rank (ROC-AUC)
5	Neural Network + Geography	0.844	0.854	0.830	0.842	0.927	1
4	Random Forest + Geography	0.827	0.834	0.816	0.825	0.910	2
3	Neural Network	0.778	0.759	0.814	0.786	0.863	3
2	Random Forest	0.758	0.754	0.766	0.760	0.837	4
1	Gradient Boosting	0.734	0.733	0.738	0.735	0.800	5
0	Logistic Regression	0.695	0.695	0.687	0.692	0.751	6

Figure 1: Comparison Bar chart of all supervised models' ROC_AUC values



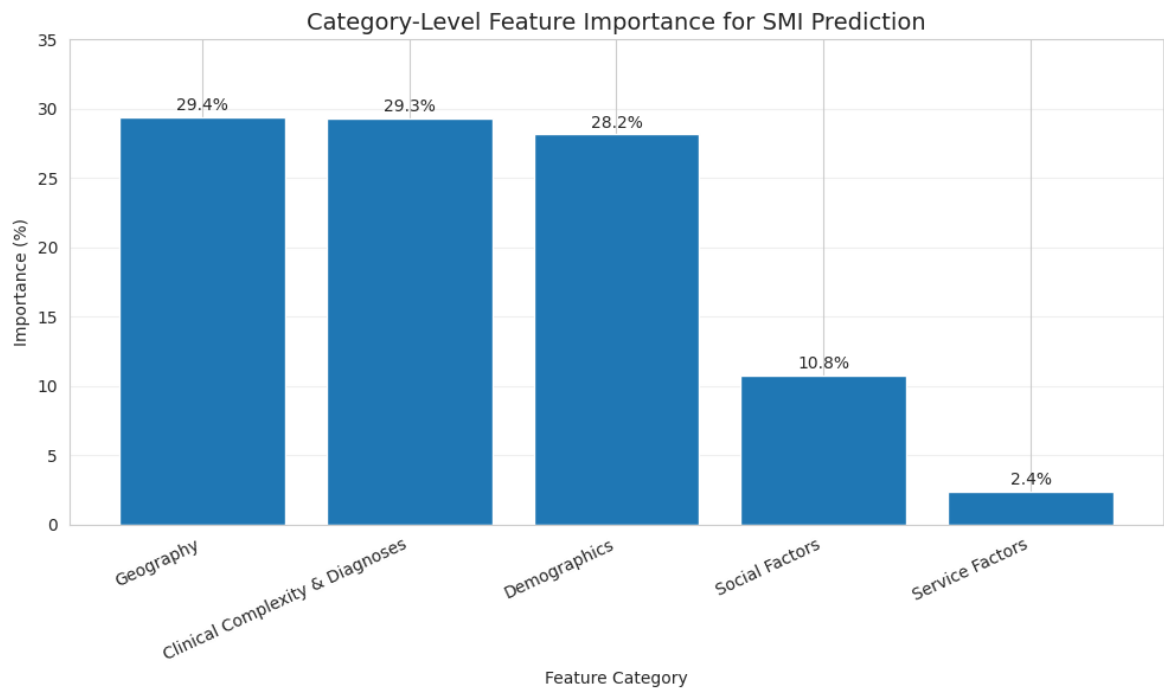
Reviewing the confusion matrix for the outstanding model (Neural Network + geography), the model correctly identified 16,603 clients with SMI, while it missed 3,397 true SMI prediction cases. This means the model correctly identified about 83% of clients with SMI cases, with a precision rate of 85%. However, the 17% missed SMI cases are also important and non-negligible, especially in healthcare. If the model is implemented, continued in-depth screening of patients in session, in addition to using predictive analytics, would minimize the impact of missed predictions. Overall, this model's performance suggests that it could serve as an effective decision support tool in identifying high-risk clients in behavioral health. See Table 2 for details of the model's true/false positives and negatives.

Table 2: Confusion Matrix for Neural Network with Geography Model

<pre>1 # Confusion Matrix 2 3 print(confusion_matrix(y_test, y_pred_nn)) 4 5 print(classification_report(y_test, y_pred_nn))</pre>					
[[17171 2829] [3397 16603]]					
		precision	recall	f1-score	support
	0	0.83	0.86	0.85	20000
	1	0.85	0.83	0.84	20000
	accuracy			0.84	40000
	macro avg	0.84	0.84	0.84	40000
	weighted avg	0.84	0.84	0.84	40000

Additionally, the feature importance analysis showed that geography, clinical complexity and diagnosis, and demographics contributed nearly equal predictive value. This suggests that SMI classification of clients is influenced not only by their individual diagnosis but also by their demographic profile and the location where they receive services. See Figure 2 below for details.

Figure 2: Category-Level Feature Importance Classification for SMI Prediction



Furthermore, the K-Means clustering analysis (unsupervised) identified two distinct primary client profiles. In cluster 0, approximately 6.8% of the client population were characterized by substantially higher rates of Serious Mental Illness (SMI), Schizophrenia, Bipolar disorder, Substance use disorder, Homelessness, Unemployment, and residence in institutional or non-

private living arrangements. This cluster also contained a higher proportion of veterans and demonstrated greater utilization of intensive behavioral health services. Given its combination of severe psychiatric conditions, substance use challenges, and social instability. This cluster is labeled the Severe Behavioral Health Complexity. Also, in cluster 1, approximately 93.2% of the client population were characterized by higher rates of Anxiety disorders, Depressive disorders, Trauma-related disorders, and ADHD. Clients in this segment were more likely to live in private residences, experience lower levels of homelessness and Unemployment, and exhibit lower rates of Schizophrenia and Substance use disorder. This cluster is labeled the Internalizing Disorders, reflecting a population whose behavioral health needs are primarily driven by mood, anxiety, and trauma-related conditions rather than psychotic-spectrum disorders.

A k-5 exploratory clustering model further revealed clinically meaningful subgroups, including anxiety complexity, substance use, personality disorder complexity, and psychotic-spectrum clusters. However, the k-2 cluster appears to be superior, as evidenced by the silhouette plot, yielding two interpretable behavioral health clusters. See Tables 3 and 4 for cluster size and SMI rate per cluster.

Table 3: K-2 Cluster size

Cluster	Count	Percent
0	13689	6.8445
1	186311	93.1555

Table 4: K-2 SMI rate per cluster

Cluster	count	mean	SMI_Rate_Percent
0	13689	0.642998	64.299803
1	186311	0.489493	48.949337

Overall, this project demonstrates the effectiveness of machine learning within behavioral health and how it can support mental health risk prediction, client segmentation, enhance service planning, and be pivotal in future digital mental health product development. Both supervised and unsupervised models revealed key findings that are meaningful and could be pivotal in improving how funded mental health care is approached across the United States. See Table 5 for the full model comparison results of the supervised and unsupervised machine learning models.

Table 5: Full Comparison of Supervised and Unsupervised Models

	Model	Learning Type	Primary Purpose	Primary Metric	Metric Value	Key Finding	
0	Logistic Regression	Supervised	SMI prediction	ROC-AUC	0.751	Baseline linear model	
1	Gradient Boosting	Supervised	SMI prediction	ROC-AUC	0.800	Improved nonlinear performance	
2	Random Forest	Supervised	SMI prediction	ROC-AUC	0.837	Strong tree-based performance	
3	Neural Network	Supervised	SMI prediction	ROC-AUC	0.863	Best model before geography	
4	Random Forest + Geography	Supervised	SMI prediction	ROC-AUC	0.910	Geography improved performance	
5	Neural Network + Geography	Supervised	SMI prediction	ROC-AUC	0.927	Best overall predictive model	
6	K-Means (K=2)	Unsupervised	Client segmentation	Silhouette Score	0.393	Best primary segmentation solution	
7	K-Means (K=5)	Unsupervised	Exploratory client segmentation	Silhouette Score	0.131	Clinically meaningful exploratory subgroups	

Strategic Recommendations

The Neural Network with Geography model demonstrated the strongest predictive performance overall, for identifying clients at elevated risks of a Serious Mental illness (SMI) diagnosis, suggesting that geography may be a meaningful driver of risk. In addition, cluster analysis revealed distinct client segments with varying levels of intervention needs and urgency. Therefore, these findings support the use of predictive analytics and segmentation to improve early identification, resource allocation, and intervention planning as highlighted below.

1. Deploy Predictive Risk Scoring
 - Implement the Neural Network model as a decision support scoring tool to identify clients at elevated risks of an SMI diagnosis and prioritize early intervention efforts.
2. Leverage Geographic Insights
 - Further explore geographic influences to identify service gaps, reporting disparities, and opportunities for regional intervention strategies and resource planning.
3. Implement Segment specific Intervention Pathways
 - Use cluster profiles to tailor intervention and care pathways instead of relying on one-size fits all approach, improving resource allocation, utilization, and client outcomes.
4. Develop scalable digital Interventions
 - Use cluster insights (client segment) to design targeted digital resources such as screening tools, mental health education modules, and early intervention support programs.

Final implementation and deployment of this project in the real-world should include further fairness checks, external validation, review by licensed mental health professionals, and business operations stakeholders.

Strategic Impact

This project demonstrated the successful application of machine learning to predict serious mental illness (SMI) using the mental health client-level data (MH_CLD), while also identifying distinct and meaningful behavioral health client segments. The findings highlight how predictive analytics can support data driven decision making by enabling earlier detection of high-risk individuals and population segments requiring targeted intervention.

The project also highlighted opportunities to improve resource allocation, intervention planning, and service delivery through risk stratification and behavioral segmentation. By supporting a more proactive and personalized approach to care, this framework provides a scalable foundation for improving behavioral health outcomes, optimizing operational resources, and informing future program and policy decisions.